A reusable Bayesian IRT workflow for small-sample instrument validation: A tutorial with a case study in teacher belief measurement

Chi Zhang[1] and Maria Pampaka[1,2]

[1]Manchester Institute of Education, The University of Manchester

[2]Department of Social Statistics, The University of Manchester

Author Note

Chi Zhang https://orcid.org/0000-0002-9535-6186

Maria Pampaka https://orcid.org/0000-0001-5481-1560

Correspondence concerning this article should be addressed to Chi Zhang, Manchester Institute of Education, The University of Manchester, Oxford Road, Manchester, England M13 9PL, Email: chi.zhang-3@manchester.ac.uk

Abstract

Bayesian Item Response Theory (IRT) offers distinct advantages for fitting complex measurement models under sample-size constraints, yet practical guidance on implementing complete Bayesian IRT workflows for applied researchers remains limited. This tutorial presents a reusable workflow for validating multidimensional instruments with limited samples using the R package brms. The workflow guides researchers through five integrated steps: (1) preprocessing decisions for sparse ordinal responses, (2) comparing competing dimensional structures including bi-factor models, (3) using discrimination parameters as diagnostic tools for identifying threshold anomalies and misfitting items, (4) selecting among ordinal link functions using cross-validation and stability diagnostics, and (5) incorporating covariates and differential item functioning analysis. Throughout, we emphasise how the rich diagnostic information provided by Bayesian posterior distributions supports informed qualitative judgment at each decision point. The workflow is illustrated through the validation of a 24-item instrument measuring Chinese secondary mathematics teachers' beliefs about equity (n = 155). The iterative process identified misfitting items, resolved threshold anomalies, and ultimately supported a bi-factor graded response model with a general inclusive belief factor and three domain-specific components. All analysis code and data are available in a companion repository.

Keywords: Bayesian item response theory; multidimensional IRT; instrument validation; Bayesian workflow; brms

A reusable Bayesian IRT workflow for small-sample instrument validation: A tutorial with a case study in teacher belief measurement

Results

Bayesian IRT Framework

The current study evaluates mathematics teachers' beliefs about equity through a Bayesian Item Response Theory (IRT) framework. As noted by Bürkner et al. (2019), Bayesian methods are particularly robust for complex multidimensional structures when dealing with limited sample sizes. In this model, the probability of teacher i endorsing item j is defined by the following two-parameter logistic (2PL) function:

$$P(y_{ij} = 1|\theta_i, a_j, b_j) = \frac{\exp[a_j(\theta_i - b_j)]}{1 + \exp[a_j(\theta_i - b_j)]}$$

Where θ_i represents the latent equity belief of teacher i, a_j is the item discrimination, and b_j is the item difficulty threshold. Following the scale development procedures described by Hinckley et al. (2026), we compared three competing structures.

```
# --- Bayesian IRT Model Specification ---
library(brms)

# Define the 2PL model formula
bf_2pl <- bf(
  ans ~ 1 + (1 | item) + (1 | person),
  disc ~ 1 + (1 | item)
)
```

Model Comparison and Validation

The results suggest that the bi-factor model provides the most parsimonious fit for measuring equity beliefs (see also Hinckley et al., 2026, for a discussion on multidimensionality). In the bi-factor specification, the latent response is determined by both a general factor (θ_G) and specific dimensions (θ_S):

$$\text{logit}[P(y_{ij} = 1)] = \alpha_{jG}\theta_{iG} + \alpha_{jS}\theta_{iS} - \beta_j$$

Consistent with the recommendations of Bürkner (2024), we implemented the models using weakly informative priors to ensure the stability of the posterior estimates. All models were checked for convergence using the R-hat diagnostic ($\hat{R} < 1.01$).

References

Bürkner, P.-C. (2024). *The brms Book: Applied Bayesian Regression Modelling Using R and Stan (Early Draft)*.

Bürkner, P.-C., Schulte, N., & Holling, H. (2019). On the Statistical and Practical Limitations of Thurstonian IRT Models. *Educational and Psychological Measurement*, *79*(5), 827–854. https://doi.org/10.1177/0013164419832063

Hinckley, K., Alvarado, M., McCoy, D. C., Jäggi, L., Dormal, M., Mäusezahl, D., Hartinger, S. M., Castellanos, A., Huaylinos Bustamante, M. L., Aguilar, L., Gastiaburú Cabello, M. C., Quispe Roncal, N. G., & Fink, G. (2026). Development and validation of the Nurturing Parenting Beliefs and Behaviors Scale (NPBBS): Measuring parenting in high Andean Peru. *Developmental Psychology*, *62*(1), 233–253. https://doi.org/10.1037/dev0002037